CS683 PA2: Summary

Task 1: Stride Pattern Prefetcher (L1D); Task 3: SPP + prefetch-aware Dynamic I-RRIP

Task 1A: IP-stride (baseline prefetcher)

`ip_stride.l1d_pref` keeps 24 IP trackers (IP, last line address, last stride, LRU). On an access by a tracked IP it computes the stride from the last line; if it equals the stored stride it prefetches the next 3 strides ahead (same page only), otherwise it just stores the new stride. It therefore captures only a *constant* stride and is blind to repeating non-uniform sequences such as $(1, 2, 3)^n$. It is tracked by a fully-associative LRU table of 24 entries, so IPs beyond 24 thrash it.

Task 1B: SPP design

- **IP table** (256 entries, direct mapped, 8-bit tag): last line (low 16 bits), last stride, a 2-bit uniform-stride confidence, and a signature of the last 3 strides (7 bits each).
- **Pattern table** (256 entries) indexed by a hash of the 3-stride signature → (next stride, 2-bit confidence). A **fallback table** (64 entries) is keyed by the last 2 strides and is used when the 3-stride entry is not confident.
- **Prediction**: if the uniform confidence is high the constant stride is used; otherwise the pattern table is walked, feeding each predicted stride back into the signature, up to the current degree. Prefetching stops at the 4KB page boundary and when the prefetch queue is full. Only loads/RFOs train. Prefetches fill L1D unless the MSHR is more than half full, then L2.
- **Accuracy throttling**: a 64-entry filter of recently filled prefetch lines; a demand hit on it counts as useful. Every 256 prefetch fills accuracy is computed and the degree set: ≥75%→6, ≥50%→4, ≥25%→2, else 1 (initial 4).

Results (25M warmup + 25M simulated instructions)

Speedup = IPC / baseline IPC (mean = geometric). Coverage = (baseline L1D load misses − misses with prefetcher)/baseline. Accuracy is the simulator's (useful / prefetches issued to the lower level). **IPC**

Config	Trace1	Trace2	Trace3	Mean
No prefetcher	0.2323	0.2512	0.3821	0.2885
IP-stride	0.2335	0.2513	0.3829	0.2892
SPP (ours)	0.2343	0.2549	0.3872	0.2921

Speedup

Config	Trace1	Trace2	Trace3	Mean
No prefetcher	1.0000	1.0000	1.0000	1.0000
IP-stride	1.0048	1.0004	1.0021	1.0024
SPP (ours)	1.0082	1.0149	1.0135	1.0122

L1D load MPKI

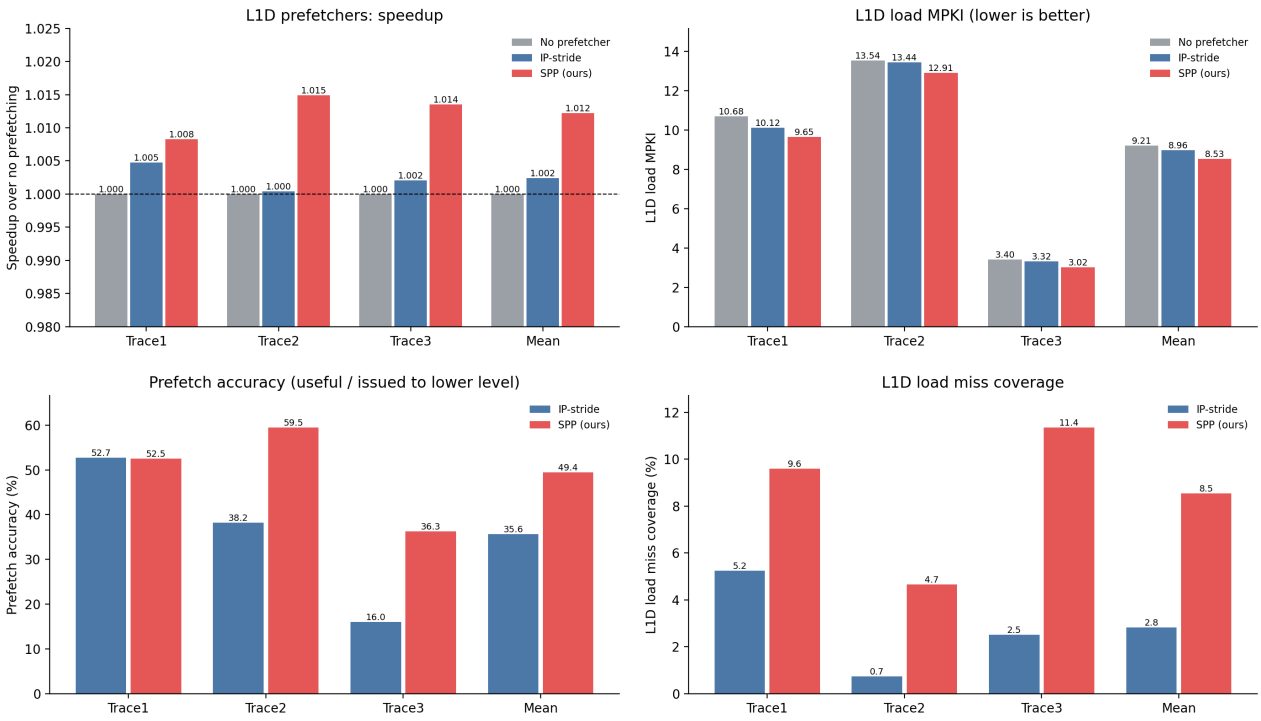
Config	Trace1	Trace2	Trace3	Mean
No prefetcher	10.678	13.538	3.403	9.206
IP-stride	10.119	13.438	3.318	8.958
SPP (ours)	9.653	12.907	3.017	8.526

Prefetch accuracy (%)

Config	Trace1	Trace2	Trace3	Mean
No prefetcher	n/a	n/a	n/a	n/a
IP-stride	52.71	38.18	15.99	35.63
SPP (ours)	52.49	59.53	36.27	49.43

L1D load coverage (%)

Config	Trace1	Trace2	Trace3	Mean
No prefetcher	n/a	n/a	n/a	n/a
IP-stride	5.24	0.74	2.51	2.83
SPP (ours)	9.60	4.66	11.36	8.54



SPP beats IP-stride on every trace in speedup and coverage (mean speedup 1.0122 vs 1.0024; coverage 8.5% vs 2.8%), and has higher accuracy on Trace 2 and 3 (tied on Trace 1). Gains are small in absolute terms: a miss breakdown showed most L1D misses come from IPs not in the table, strides beyond a page (forbidden by the spec) or irregular accesses; only about 3 to 5% of misses occur while a stride pattern is active.

Storage overhead (exact)

Structure	Entries	Bits per entry	Total bits
IP table	256	valid 1 + tag 8 + last line 16 + last stride 7 + uni conf 2 + 3×7 signature + hist count 2 = 57	14592
Pattern table	256	stride 7 + conf 2 = 9	2304
Fallback table	64	stride 7 + conf 2 = 9	576
Recent-prefetch filter	64	valid 1 + tag 10 = 11	704
Counters	n/a	issued 9 + useful 9 + degree 3	21
Total		2274.6 B ≈ 2.2 KB	**18197**

For comparison, the provided IP-stride uses 24 × (64+64+64+32) = 5376 bits as coded; SPP uses about 3.4× more storage.

Threshold and degree rationale

Accuracy thresholds 75/50/25% divide accuracy into four bands: above 75% prefetches are almost always used, so aggressive depth (6) is cheap; 50 to 75% is the normal band (4); 25 to 50% means over half the prefetches are wasted, so depth is halved (2); below 25% prefetching costs more bandwidth and L1D pollution than it saves, so only one line is issued. The values are round quartile boundaries; I validated them by sweeping: thresholds 50/30/10, 60/40/20, 90/70/40 and degree sets (8,4,2,1), (8,6,4,2) all lie within ±0.1% of mean speedup (1.0115 to 1.0129), so the result is insensitive to them. Fixed degrees 1/2/4/6 give mean speedups 1.0076/1.0091/1.0101/1.0102, so deeper is better up to 4, and flat beyond. **Honest finding:** dynamic throttling gives the same speedup as fixed degree 4 (1.0122 vs 1.0124) but 4 to 9 points higher accuracy, i.e. it saves bandwidth without losing performance.

Ablations and design validation

Configuration	Mean speedup	Note
Uniform stride only (no pattern table)	1.0043	≈ IP-stride-like
Pattern table, no fallback	1.0100	
Pattern table + fallback (final)	1.0122	

IP-table size (without fallback): 64/128/256/512/1024 entries → 1.0054/1.0078/1.0100/1.0107/1.0111 (6.7/10.3/17.6/32.2/61.4 Kbit); 256 chosen as the knee. A single 2-stride table alone reached 1.0126, within noise of the final hybrid; the hybrid was kept because the 3-stride signature disambiguates longer patterns. Synthetic tests (uniform, non-uniform, phase changes, random, interleaved IPs) give about 97% coverage on regular streams, about 0% on random, and 0 page-boundary violations in 186K prefetches; the one weakness is interleaved IPs sharing a signature in the pattern table (78%), a known trade-off since IP bits in the index lowered real-trace speedup.

Task 3: SPP with prefetch-aware Dynamic I-RRIP

Policy (`replacement/dyn_irrip.l2c_repl`, applied in I-RRIP mode only; the RRIP leader sets are unchanged so set dueling still compares against plain RRIP). A prefetched data line is inserted at RRPV 3 (first eviction candidate) instead of 2, and a prefetch request that hits a resident line does not promote it. Instruction lines are unchanged (insert/promote to 0); demand hits on data still promote by one step, so a prefetched line that is actually used moves up. Eviction is unchanged.

Rationale. SPP brings lines into L1D, so the L2 copy of a prefetched line is a duplicate that is only needed if the L1D copy is evicted before use; a speculative line should therefore not displace demand data or instruction lines, which the L2 is supposed to protect. A prefetch hit is not evidence of reuse (the prefetcher re-requests a line it already fetched), so promoting on it would keep stale lines alive. Inserting at the distant position lets a useless

prefetch leave quickly, while a demand hit still rescues a useful one. The insertion RRPV was swept (1/2/3); distant insertion is best on mean speedup.

Results (25M warmup + 25M simulation instructions; speedup is geometric mean over the three traces). **IPC**

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	0.2323	0.2512	0.3821	0.2885
SPP	0.2343	0.2549	0.3872	0.2921
Dynamic IRRIP	0.2323	0.2525	0.3821	0.2889
SPP + Dyn IRRIP (pf-unaware)	0.2342	0.2562	0.3874	0.2926
SPP + Dyn IRRIP (pf-aware)	0.2344	0.2564	0.3875	0.2928
PF insert RRPV 1	0.2342	0.2557	0.3872	0.2923
PF insert RRPV 2	0.2344	0.2561	0.3874	0.2926
PF insert RRPV 3 (final)	0.2344	0.2564	0.3875	0.2928

Speedup over baseline

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	1.0000	1.0000	1.0000	1.0000
SPP	1.0082	1.0149	1.0135	1.0122
Dynamic IRRIP	0.9997	1.0052	1.0000	1.0017
SPP + Dyn IRRIP (pf-unaware)	1.0078	1.0200	1.0139	1.0139
SPP + Dyn IRRIP (pf-aware)	1.0087	1.0208	1.0143	1.0146
PF insert RRPV 1	1.0079	1.0179	1.0134	1.0131
PF insert RRPV 2	1.0087	1.0197	1.0140	1.0141
PF insert RRPV 3 (final)	1.0087	1.0208	1.0143	1.0146

L1D load MPKI

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	10.678	13.538	3.403	9.206
SPP	9.653	12.907	3.017	8.526
Dynamic IRRIP	10.678	13.538	3.403	9.207
SPP + Dyn IRRIP (pf-unaware)	9.657	12.906	3.017	8.526
SPP + Dyn IRRIP (pf-aware)	9.661	12.906	3.000	8.523
PF insert RRPV 1	9.657	12.904	3.025	8.529
PF insert RRPV 2	9.656	12.904	3.014	8.525
PF insert RRPV 3 (final)	9.661	12.906	3.000	8.523

L1D prefetch accuracy (%)

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	n/a	n/a	n/a	n/a
SPP	52.49	59.53	36.27	49.43
Dynamic IRRIP	n/a	n/a	n/a	n/a
SPP + Dyn IRRIP (pf-unaware)	52.52	59.68	36.21	49.47
SPP + Dyn IRRIP (pf-aware)	52.08	59.61	36.94	49.54
PF insert RRPV 1	52.03	59.68	35.92	49.21
PF insert RRPV 2	52.35	59.72	36.36	49.48
PF insert RRPV 3 (final)	52.08	59.61	36.94	49.54

L1D load coverage (%)

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	0.00	0.00	0.00	0.00
SPP	9.60	4.66	11.36	8.54
Dynamic IRRIP	0.00	-0.00	0.00	-0.00
SPP + Dyn IRRIP (pf-unaware)	9.57	4.67	11.36	8.53
SPP + Dyn IRRIP (pf-aware)	9.52	4.67	11.84	8.68
PF insert RRPV 1	9.56	4.68	11.11	8.45
PF insert RRPV 2	9.57	4.68	11.45	8.57
PF insert RRPV 3 (final)	9.52	4.67	11.84	8.68

L2C load MPKI

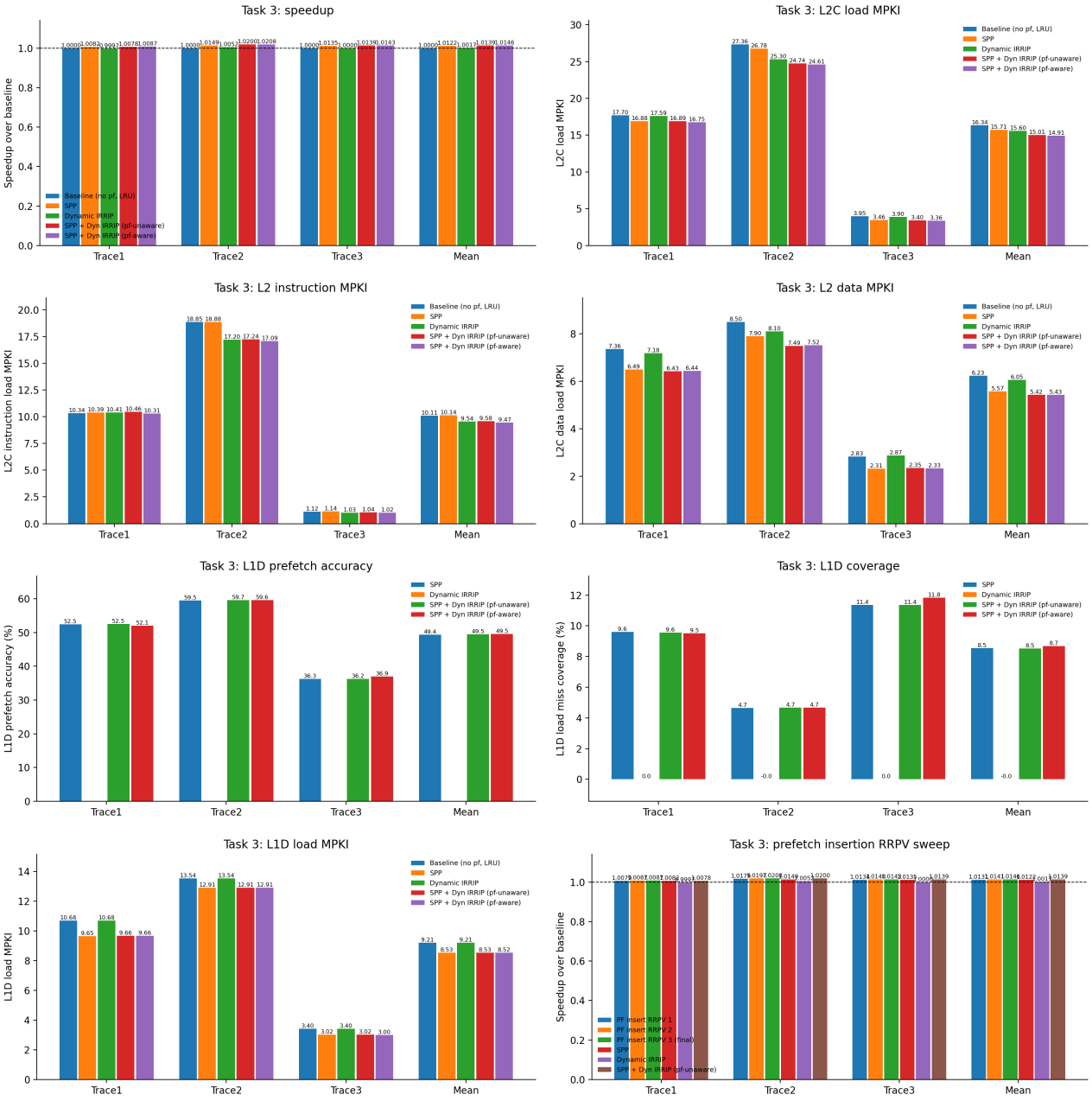
Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	17.696	27.357	3.954	16.336
SPP	16.880	26.784	3.456	15.707
Dynamic IRRIP	17.590	25.298	3.899	15.595
SPP + Dyn IRRIP (pf-unaware)	16.891	24.736	3.396	15.008
SPP + Dyn IRRIP (pf-aware)	16.749	24.609	3.359	14.906
PF insert RRPV 1	16.905	25.093	3.422	15.140
PF insert RRPV 2	16.767	24.779	3.388	14.978
PF insert RRPV 3 (final)	16.749	24.609	3.359	14.906

L2C instruction load MPKI

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	10.338	18.853	1.124	10.105
SPP	10.389	18.882	1.144	10.138
Dynamic IRRIP	10.408	17.195	1.029	9.544
SPP + Dyn IRRIP (pf-unaware)	10.464	17.244	1.043	9.583
SPP + Dyn IRRIP (pf-aware)	10.306	17.088	1.024	9.472
PF insert RRPV 1	10.475	17.677	1.055	9.736
PF insert RRPV 2	10.335	17.292	1.042	9.556
PF insert RRPV 3 (final)	10.306	17.088	1.024	9.472

L2C data load MPKI

Config	Trace1	Trace2	Trace3	Mean
Baseline (no pf, LRU)	7.358	8.504	2.830	6.231
SPP	6.491	7.903	2.312	5.569
Dynamic IRRIP	7.181	8.102	2.869	6.051
SPP + Dyn IRRIP (pf-unaware)	6.427	7.493	2.353	5.424
SPP + Dyn IRRIP (pf-aware)	6.443	7.522	2.335	5.433
PF insert RRPV 1	6.431	7.417	2.367	5.405
PF insert RRPV 2	6.431	7.487	2.346	5.422
PF insert RRPV 3 (final)	6.443	7.522	2.335	5.433



Findings. SPP + prefetch-aware Dyn I-RRIP reaches mean speedup 1.0146, against 1.0122 for SPP alone, 1.0017 for Dynamic I-RRIP alone and 1.0139 for SPP with the prefetch-unaware Dynamic I-RRIP; its IPC is highest on all three traces (0.2344/0.2564/0.3875), so it outperforms both SPP and Dynamic I-RRIP in IPC. The margins are small: on Trace 1 the gain over SPP is ≈0.0001 IPC, and the gain over the prefetch-unaware policy is 0.0007 in mean speedup, within noise for a single 25M-instruction window. The effect is clearest at L2: mean L2 load MPKI falls from 15.01 (unaware) to 14.91, and L2 instruction load MPKI from 9.58 to 9.47, because prefetch fills no longer displace instruction lines. L2 data load MPKI is slightly higher than unaware (5.43 vs 5.42), and prefetch accuracy and coverage at L1D are essentially unchanged (49.5% / 8.7%), since the change affects L2 residency, not what is prefetched.

Limitations

One 25M-instruction window per trace (the simulator is deterministic, but differences of ∼0.1% may not generalise). ChampSim's warmup skips front-end stalls, so the measured region starts with caches and tables relatively cold;

this affects all configurations equally. The accuracy counter does not track prefetches that fill L2.